

CoVeR: Coverage-Based Routing of Verifier Calls in Agentic Retrieval

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Abstract

An agentic retrieval system issues a sequence of search queries and must decide, at each step, whether the evidence collected so far is enough to stop. Delegating that decision to an LLM verifier or a prompt judge makes stopping reliable, but the verifier then reprocesses the growing evidence after every retrieval step, a substantial repeated cost. We show that most of these calls can be skipped without materially changing answer accuracy: a single threshold on a frozen sentence-embedding *coverage margin* detects the states in which the evidence is still plainly incomplete, and the verifier is called only on the ambiguous remainder — a gate we call CoVeR (**C**overage-based **V**erifier **R**outing). Across three multi-hop QA benchmarks, with the evaluation protocol fixed before the full-scale run, the CoVeR-gated agent matches the answer accuracy of both the full-budget agent and the always-verify baseline within a fraction of an EM point. It cuts 62–68% of verifier calls, and 93% in a saturated regime. Routers built on evidence counts, lexical overlap, or BM25 relevance, alone or learned in combination, give weaker overall trade-offs, the gate transfers without re-tuning across deciders and agent scales, and its drafter distills into a 921k-parameter head atop the frozen encoder, leaving no LLM in the routing loop. The same signal cannot *replace* verification: matching a claim is far easier than deciding the claim is supported.

1 Introduction

A retrieval-augmented search agent answers a multi-hop question by issuing a sequence of queries, each retrieving new evidence, until it decides it has enough to answer (Lewis et al. 2020; Press et al. 2023; Yao et al. 2023; Jin et al. 2025; Chen et al. 2025). When to stop is a control problem: stopping early can miss a required reasoning hop, while continuing past sufficiency wastes retrieval calls, adds latency, and buries the answer in distracting context. A reliable way to make the decision is to frame stopping as *evidence coverage* and delegate it to a *decider* that reads the accumulated evidence and rules on whether each required claim is now supported, either with a trained coverage verifier or with a zero-shot sufficiency judge (Yang et al. 2025; Park, Cho, and Lee 2025; Asai et al. 2024). The catch is that the decider is

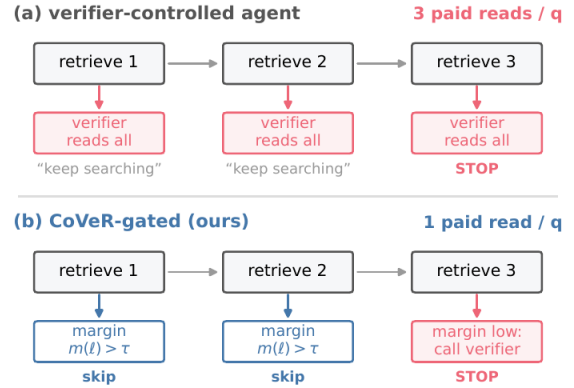


Figure 1: **Routing the verification bill.** (a) The verifier reads all evidence at every step. (b) CoVeR pays only for the ambiguous step (outcome preservation: §5.1).

itself an LLM, and it is consulted after every retrieval step. Once the stopping policy is fixed, repeatedly invoking the decider becomes an avoidable verification overhead: in our setting, nine to eleven claim verdicts per question, each a full read of the accumulated evidence (§5.5).

Much of that reading is spent on states whose verdict is not in doubt. Consider a 2WikiMultiHopQA question: “Which film has the director died earlier, *Remember the Day* or *Cast Up by the Sea*?” The agent needs each film’s director and death date. Its first two steps retrieve *Remember the Day*’s side (Henry King, d. 1982), but nothing yet links *Cast Up by the Sea* to any director, and a verifier-controlled run pays for two “keep searching” verdicts (Figure 1). Both were predictable without an LLM read: the least-covered claim’s margin holds at 0.18 against a threshold of 0.16, so CoVeR skips both calls. At the third step the missing link arrives (“*Cast Up by the Sea* is a 1916 Australian film directed by John Gavin”, d. 1938), the margin drops to 0.13, the gate pays for one verdict, the verifier fires — and every arm returns the correct answer (the full trajectory, with qid, is in Appendix A of the supplementary material).

Our method operationalizes this observation. Embed each expected hop claim and each retrieved sentence with a frozen encoder, and define a per-claim *coverage margin* as one mi-

nus the best cosine similarity to any evidence retrieved so far; the state margin is the largest per-claim margin. A single threshold τ routes the decider: while the least-covered claim is still clearly uncovered, the verdict is obviously “keep searching” and we skip the call; otherwise we pay for a verdict and follow it. The check itself is a handful of cached dot products, negligible next to a decider call. A skip cannot introduce an unverified stop; it can only defer stopping to a later, adjudicated loop. Deferral can still change what the agent reads and hence its answer, an effect our evaluation measures directly (§5.1), but every stop still passes through the LLM decider.

We organize the paper around two questions. Can the margin simply *replace* the decider? No. With oracle evidence targets it separates covered from uncovered claims well, but the deployable version built on generated claim strings falls far short of that ceiling, a label-free predictor trained to close the gap fails, and a margin-only stopping rule forced to match the verifier’s safety degenerates to never stopping (§5.2). Deciding that evidence supports a claim requires reading how entities and relations compose; cosine proximity does not establish it. This is the matching-is-not-supporting gap familiar from natural-language inference and attribution evaluation (Thorne et al. 2018; Bohnet et al. 2022; Gao et al. 2023), measured here for coverage stopping.

Can the margin *route* the decider? Yes, and this is the main result. With thresholds selected leave-one-dataset-out, the CoVeR-gated agent matches the accuracy of the full-budget agent and the always-verify baseline — every 95% CI on the EM difference lies within ± 1.2 pp — at 62–68% fewer verifier calls and 72–97% fewer verifier input tokens (§5.1, §5.5). The saving is not available to cheaper signals: routers built on evidence counts, lexical overlap, or BM25 relevance cannot reach the same call cut at the same safety bar (§5.3). Nor is it tied to one decider or one scale: the same threshold, with no re-tuning, routes a public prompt judge and a 7B agent and passes an initial open-corpus check, and the gate-side claim embeddings distill into a sub-1M drafter that leaves no LLM in the routing path (§5.4).

We further analyze whether the routing signal can also veto erroneous decider stops; the observed gains are directionally consistent but not statistically conclusive. The margin vetoes a volatile prompt judge’s premature stops, a consistent but not statistically significant gain; it cannot audit the trained verifier, whose false stops are caused by the same deceptive similarity that fools the margin itself (§6).

Our contributions are summarized as follows:

- **One-sided verifier routing.** A regime that skips only clearly futile verifier calls while every stopping decision remains decider-issued — viable because matching a claim is easier than judging it supported (§5.2).
- **CoVeR.** An instantiation by a single threshold on a frozen embedding coverage margin, selected leave-one-dataset-out and never on the target dataset: it cuts 62–68% of verifier calls at EM within a fraction of a point of both references, and no cheaper or learned router we test matches this trade-off (§5.1–§5.3).
- **Transfer and limits.** The same gate routes three deciders

and a larger agent without re-tuning, and distills into a 921k-parameter drafter whose claim-free pairing removes claim generation entirely (§5.4); but the margin cannot replace verification, and it cannot correct errors it shares with the verifier (§6).

2 Background and Related Work

Learning to defer, cascades, and confidence-gated routing.

Our gate is a cascade that invokes an expensive expert only when a cheap stage is uncertain (Chen, Zaharia, and Zou 2024); unlike model cascades, whose stages differ in size, ours differ in *kind* (a geometric check versus a reading judgment), and it repeats at every retrieval step. From learning to defer we take one licensing point: the rejector need not be accurate on the decision itself, only calibrated about when it is confident (Madras, Pitassi, and Zemel 2018; Mozannar and Sontag 2020) — which is why a signal too blunt to decide stopping (§5.2) can still route it. Similar asymmetric gates appear in SAGE, which defers only ambiguous memory writes to an LLM (Wang, Brahma, and Henao 2026). Closest is selective verification for reasoning (Qu 2026), which allocates verifier calls among candidate reasoning states; routers pick a retrieval strategy per question (Li, Yan, and Käfer 2026) or prune retrieval depth by question difficulty (Java et al. 2026). CoVeR routes neither: it gates *when a fixed stopping decider is consulted*, one-sidedly, never issuing an unverified stop.

Matching is not supporting. Whether retrieved text *supports* a claim is the object of NLI and attribution evaluation: FEVER casts verification as entailment over retrieved sentences (Thorne et al. 2018), and attribution metrics score whether a statement is *entailed* by its cited evidence rather than merely similar to it (Bohnet et al. 2022; Gao et al. 2023); Joren et al. (2025) lifts this to *sufficiency*. Our replacement negative measures the same gap in embedding space for per-hop coverage stopping: cosine proximity is a good difficulty signal but a poor sufficiency decision.

Speculative execution. Speculative decoding drafts tokens for a large model to verify in parallel, exactly preserving the output (Leviathan, Kalman, and Matias 2023; Chen et al. 2023); Speculative RAG drafts *answers* (Wang et al. 2025). There, verification is one parallel forward pass and every draft is checked; a skipped verifier call has no parallel check, so our guarantee is statistical, not exact.

Adaptive stopping and retrieval. Prior work learns or defines the retrieval control action: triggering on decoding-time uncertainty (Jiang et al. 2023) or question complexity (Jeong et al. 2024), and, closer to stopping, ending on an answer span (Trivedi et al. 2023), a trained sufficiency critic (Yang et al. 2025), or a value-based stop policy (Park, Cho, and Lee 2025). All decide the control *action*; we take the policy as given and lower the cost of delivering it, composably with any of them. To our knowledge, no prior work routes verifier calls with a one-sided frozen coverage signal while keeping every stop decider-issued.

3 Method

Setup. A search agent runs loops $\ell = 1, \dots, L$ ($L=6$ in our data), accumulating evidence S_ℓ . The question induces a small set of expected hop *claims* $\{c_1, \dots, c_k\}$ ($k \approx 2$). A *decider* D (a trained verifier or a prompt judge) reads S_ℓ each loop and rules on whether every claim is covered; the agent stops at the first loop D rules the state covered (we say D *fires*). D is invoked once per loop until it fires — the cost we route.

Frozen-embedding coverage margin. Let ϕ be a frozen E5 sentence encoder (Wang et al. 2022). For claim c_j with target text t_j and evidence S_ℓ , define the per-claim coverage margin and state margin

$$m_j(\ell) = 1 - \max_{s \in S_\ell} \cos(\phi(t_j), \phi(s)), \quad m(\ell) = \max_j m_j(\ell).$$

A high margin means “no retrieved sentence is close to this claim.” We call this score a coverage margin: a frozen similarity shortfall, not a learned energy. The target t_j is a *generated* expected-claim string (deployable, no gold labels), a gold supporting sentence (an oracle diagnostic only), or a learned predictor’s output (below). Computing $m(\ell)$ is a handful of cached dot products, negligible next to a decider call.

The CoVeR gate. Figure 2 summarizes the gate and Algorithm 1 gives the full loop: before each decider call, evaluate the state margin:

$$\begin{aligned} m(\ell) > \tau &: \text{skip the call, continue;} \\ m(\ell) \leq \tau &: \text{call } D \text{ and follow its verdict.} \end{aligned}$$

We call this rule **CoVeR**. As in the introduction, a skip cannot end a trajectory, so the rule is one-sided by construction.

One-sidedness is a design consequence, not a convenience. The margin is built to detect states that are *plainly incomplete*, not states that are *sufficient*: a large margin means no retrieved sentence comes close to some expected claim, which is cheap evidence that the search is unfinished. The converse does not follow. Establishing that evidence *supports* a claim requires reading how entities and relations compose, and cosine proximity does not establish it. We therefore trust the margin only on the side where a geometric check is reliable, and leave every stop to the decider. Section 5.2 measures both sides of this asymmetry directly. A second, lower early-exit threshold (stop-without-a-call) would trust the margin on the side it is weak and break the one-sided guarantee, so the primary system uses the single threshold; we evaluate the two-threshold variant only as an ablation (App. B of the supplement).

Threshold selection. Thresholds are selected on training splits only, leave-one-dataset-out (LODO): τ for a target dataset is chosen on the *other two* datasets’ train-600 and evaluated once on the target’s held-out full split. Selection maximizes call reduction subject to two training-side constraints: *stop-loop agreement* with always-verify (the fraction of questions stopping at the same loop) must stay at or above 0.90, and *coverage safety* (the fraction of early fired stops at or past the gold-labeled all-claims-covered loop)

Algorithm 1 The CoVeR-gated agent loop

Require: question q ; decider D ; encoder ϕ ; threshold τ ; budget L

- 1: $\{e_j\}_{j=1}^k \leftarrow \phi(\text{Claims}(q))$ or $\text{drafter}(q)$ {embedded LLM claim text, or predicted embeddings}
- 2: $S_0 \leftarrow \emptyset$
- 3: **for** $\ell = 1$ **to** L **do**
- 4: $S_\ell \leftarrow S_{\ell-1} \cup \text{retrieve}(q, \ell)$ {embed new sentences}
- 5: $m(\ell) \leftarrow \max_j (1 - \max_{s \in S_\ell} \cos(e_j, \phi(s)))$
- 6: **if** $m(\ell) > \tau$ **then**
- 7: **continue** {skip: defer, never stop}
- 8: **end if**
- 9: $v \leftarrow D(S_\ell)$ {pay for a verdict}
- 10: **if** $v = \text{all covered}$ **then**
- 11: **return** answer {stop is decider-issued}
- 12: **end if**
- 13: **end for**
- 14: **return** answer at budget exhaustion

must be at least the decider’s own. This yields $\tau=0.16$ on all three folds. The two constraints encode the one-sided design rather than a target performance level: the agreement floor bounds how far the routed agent may drift from the policy it is routing, and the coverage-safety floor forbids buying call cut by skipping the very loops the decider would have used to stop. Both bars are fixed across folds; only the threshold is searched, and only on the other datasets’ training splits. All three folds select the same value, so the operating point does not depend on which dataset is held out.

Extension: removing claim generation from the gate. The primary CoVeR system uses generated expected claims, costing one LLM claim-generation pass per question. As a deployment extension we distill those embeddings into a lightweight question-to-claim drafter. A sub-1M MLP maps the question embedding to k slot embeddings. It is trained with a cosine loss against the generated-claim embeddings, while a small classification head predicts k ; there are *no gold labels*. At inference the predicted embeddings replace the generated claim text *in the margin computation*, so the gate needs only the frozen encoder and this MLP; a claim-text decider (the trained verifier) still needs claim generation when called — as does always-verify.

Extension: intra-call ordering. When D is called, the all-covered check short-circuits at the first uncovered claim; when scoring is sequential, D may optionally evaluate claims in descending-margin order, which reaches that claim sooner — it prioritizes the claim most likely to be uncovered. Order-invariance preserves the verdict exactly; the saving arises only under sequential scoring (§5.5).

4 Experimental Setup

Datasets, agents, deciders. We evaluate on HotpotQA (distractor) (Yang et al. 2018), 2WikiMultiHopQA (2Wiki) (Ho et al. 2020), and MuSiQue (Trivedi et al. 2022). The host agent is Self-Ask (Press et al. 2023) with Qwen2.5-3B-Instruct (Yang et al. 2024) (and Qwen2.5-7B for the scale

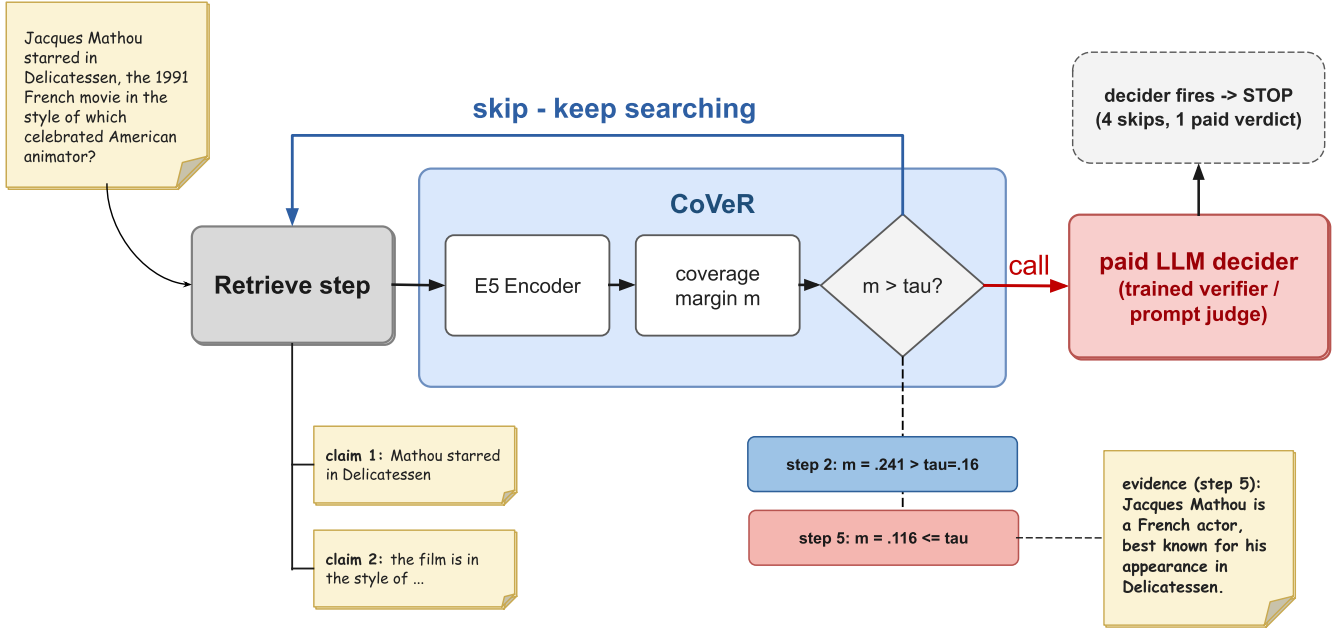


Figure 2: **CoVeR routes the decoder** (a routed HotpotQA trajectory). Claims and evidence are embedded by a frozen encoder; one threshold on the coverage margin skips the clearly-uncovered steps and pays for the one ambiguous state, where the swappable decoder rules. A skip can only defer stopping — every stop is decoder-issued.

test), maximum 6 retrieval loops (the simulator’s budget parameter of 8 was never binding), over a BM25 sentence index (Robertson and Zaragoza 2009) of gold+distractor paragraphs, top- $K=3$ per query (a closed pool isolating stopping from retrieval failure). We use *two deciders*: a public **zero-shot prompt-only sufficiency judge** (same Qwen2.5-3B, no training, stop iff $p_{\text{yes}} \geq \tau_{\text{judge}}$), and a **trained-verifier instantiation** (a Qwen2.5-3B+LoRA coverage verifier (Hu et al. 2022), described in the supplementary material and used without modification or retraining). Reporting both decouples the method from any single decoder.

Systems. We compare three systems built on identical trajectories: the **full-budget** agent runs all L retrieval loops; the **always-verify** agent invokes the decoder after every loop and stops when it fires; the **CoVeR-gated** agent invokes the decoder only when $m(\ell) \leq \tau$.

Endpoints. EM is exact match, scored as Std-Ext EM: a frozen Qwen2.5-3B extractor reads cumulative chunks at the stop loop, so all arms share the same extractor procedure. The endpoint is ΔEM of the CoVeR-gated agent against each reference: the full-budget agent (a conservative reference, not an assumed optimum) and the always-verify baseline (the stopping policy whose calls the gate routes). *Call cut* is the fraction of the always-verify run’s decoder calls that the gate skips on the same trajectories, and *stop-loop agreement* is the fraction of questions on which the gated and always-verify runs stop at the same loop. Reported together, the two endpoints separate outcome preservation from policy imitation: EM measures the answer the agent finally returns, while agreement measures how closely the routed

agent reproduces the decoder’s own stopping behavior. The main result evaluates $n=1000/1000/911$ unique questions on HotpotQA/2Wiki/MuSiQue (populations differ by protocol; transfer arms use their own splits, §5.4), and 4-bit re-extraction can flip individual questions; full accounting in Appendix E of the supplementary material.

Confidence intervals. All CIs are paired bootstrap ($10,000 \times$ main, $2,000 \times$ AUROC, seed 13; Efron and Tibshirani 1993); AUROC CIs cluster by question. With $n \leq 1000$ at this EM level, effects below ~ 1 pp are within noise on HotpotQA/2Wiki. A formal non-inferiority and equivalence analysis, with margins fixed before the full run, supports the same conclusions and is reported in Appendix B of the supplementary material.

Artifacts and protocol. The trained verifier is not introduced or trained in this paper; it is documented in the supplementary material. We use its cached trajectories, verdicts, generated claims, and extractions read-only, without modifying its training or predictions — CoVeR asks only whether a call can change the current control action. Every success bar was fixed before the corresponding full-scale run, and freshly extracted arms are parity-checked against the cache (≥ 0.98). Code and archived result sets are in the code supplement; complete provenance, parity checks, pilot analyses, and protocol history are reported in Appendix B of the supplementary material.

5 Results

Stopping verification bundles two abilities: detecting that evidence is still plainly incomplete, and judging that it has

become sufficient — and our results test whether they separate, the first served by cheap geometry, the second reserved for the decider. We report the routing result first (§5.1), then the asymmetry that licenses it (§5.2), the comparison against cheaper routers (§5.3), transfer (§5.4), and workload (§5.5).

5.1 Main Result: Answer Quality Preserved at a Fraction of the Verifier Calls

Table 1 reports the primary LODO result for CoVeR with the trained-verifier decider, evaluated on up to 1000 questions per dataset. The CoVeR-gated agent matches both references within a fraction of an EM point while cutting the majority of verifier calls.

Across the three datasets, EM differs from the full-budget and always-verify references by at most 0.3 pp, while verifier calls fall by 62.2% on HotpotQA and 67.8% on 2Wiki; every 95% CI on these differences lies within ± 1.2 pp (the full intervals, a pooled analysis, and the previously specified equivalence tests are in App. B). A post-hoc token-F1 check shows the same pattern (secondary metric; App. B). MuSiQue is a saturated regime and is interpreted separately below. The loop column reads in the gate’s favor as well: the gated agent finishes 9.8–11.9% *fewer* retrieval loops than the full-budget agent on HotpotQA/2Wiki — adjudicated stops end trajectories the budget reference would run to exhaustion — while deferring only mildly relative to always-verify (§5.5).

Three datasets, two regimes. The three rows are not three replications of one effect; they differ in how much room the decider leaves to route. HotpotQA and 2Wiki are *headroom* regimes: the verifier fires on a substantial share of loops, so the calls it receives include many whose verdict is genuinely in doubt, and skipping only the plainly incomplete states removes about two thirds of the verifier calls while preserving answer accuracy. MuSiQue is *saturated*: the verifier fires on 2.6% of loops, most trajectories reach the loop budget under every system, and the gate can skip almost every call precisely because almost none of them would have changed the control action. The saturated cell therefore tests safety rather than savings — near-total skipping does not break a decider that was not going to stop anyway — which is why we report 62–68% as the headline reduction: it is the range measured where the stopping decision was actually live.

5.2 Why the Margin Can Route but Cannot Replace

This subsection tests the design premise of §3: identifying plainly incomplete states is enough to route, but not enough to decide that a state has become sufficient. Treating each claim–state pair as a binary instance (covered or not, per the verifier’s labels) with the margin as the score, the margin reaches AUROC .89/.90/.83 with oracle targets (Table 2); the deployable generated-claim margin reaches only .69/.68/.76, a gap whose CIs do not overlap.

A label-free sub-1M predictor does not close the gap: it lands at or below the generated-claim floor on every dataset, because its only available training target — the rank-0 retrieved chunk — teaches the retrieval distribution, not which

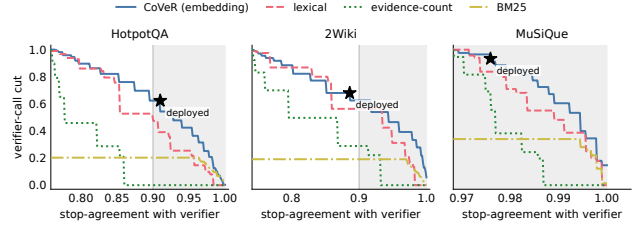


Figure 3: **Router frontier (descriptive).** Achievable verifier-call cut vs. stop-loop agreement with the always-verify decider, per dataset (envelope: max cut at agreement $\geq$ each x ; shaded: the ≥ 0.90 bar). The star marks the deployed one-threshold operating point (0.622 at 0.910, HotpotQA).

evidence is *needed*. Forced to match the verifier’s coverage-safety as a stopping rule, the margin drives loop savings to ≈ 0 and never stops on any dataset at a safety-matched operating point. Judging support requires reading how entities and relations compose (a distinction NLI and attribution evaluation long enforce (Thorne et al. 2018; Gao et al. 2023)); cosine proximity does not cross it. This is why CoVeR retains the decider for every stopping decision.

5.3 Comparison with Simpler Routing Signals

Would a cheaper signal route as well? We compare the frozen-embedding margin against three same-interface controls on a shared frontier: an *evidence-count* router (skip while few sentences are retrieved), a *lexical-overlap* router (skip while content-word recall of each claim’s tokens is low), and a *BM25* router (a per-claim Okapi-BM25 coverage margin over the question’s own pool). For each router we report the eval-set frontier: the maximum call cut holding stop-loop agreement with the always-verify decider at or above 0.90 (exact frontier values and LODO operating points: App. J).

On this frontier the embedding gives the highest feasible cut at the 0.90 bar — decisive on HotpotQA, modest on 2Wiki, a saturated tie on MuSiQue — with none of the cheaper proxies or learned combinations we test reaching a better frontier at that bar (Figure 3; learned and LODO details in App. J); Table 1’s deployed point is the LODO operating point, not this frontier. The MuSiQue tie is the saturation effect of §5.1: near-total cuts are open to any signal there, though BM25’s ordering still fails to protect the few early-fired states (App. J). Matched-budget random skipping and verifier self-gating also fall short. A random skipper calibrated to CoVeR’s exact call cut loses 6.4/7.0 pp of stop-agreement on HotpotQA/2Wiki (MuSiQue ties; the pre-specified ≥ 5 pp bar is met on two of the three datasets), so the budget alone does not buy the agreement — the margin’s choice of *which* loops to skip does. And letting the verifier gate itself from its own unmatched-claim count reaches an envelope below CoVeR’s on every dataset at the 0.90 agreement bar (App. J): self-gating pays a verifier call to obtain its signal, so it cannot skip ahead of the state it has already paid to see. The evidence-count and BM25 routers are never competitive; against the harder lexical router, the

Dataset	EM			Δ EM (pp) vs.		Call cut	Loop Δ	Agree.
	Full-budget	CoVeR	Always-verify	Full-budget	Always-verify	vs. Always	vs. Full	vs. Always
HotpotQA	0.3060	0.3040	0.3050	−0.20	−0.10	62.2%	−11.9%	.910
2Wiki	0.1840	0.1870	0.1850	+0.30	+0.20	67.8%	−9.8%	.886
MuSiQue [†]	0.0801	0.0812	0.0812	+0.11	+0.00	93.0%	−0.6%	.976

Table 1: **Routing at up to $n=1000$, LODO thresholds, trained-verifier decider.** Δ EM in pp (95% CIs in text; formal tests in App. B). Agree. = held-out stop-loop agreement (≥ 0.90 applies at LODO selection). [†]Saturated (§5.1).

Dataset	AUROC	
	Oracle	Generated-claim
HotpotQA	0.892	0.686
2Wiki	0.895	0.680
MuSiQue	0.831	0.760

Table 2: **Margin vs. per-hop coverage** ($n=1000$; CIs in App. H). A high oracle ceiling with a much weaker deployable floor: the geometry can express coverage, but matching a generated claim string cannot read it. Oracle and generated-claim targets come from different label runs (a ceiling/floor contrast, not a controlled ablation).

embedding’s cut is +10–17 pp larger on HotpotQA at every agreement bar, matched on 2Wiki only at the 0.92 bar, and tied on saturated MuSiQue — the advantage is largest where call-cut headroom exists. Routing is also not a repackaged loop-index rule: against skip-first- k and every-other-loop heuristics matched on either call cut or agreement, none matches the gate on both (HotpotQA 0.63 cut at 0.915 agreement vs. skip-first-4’s 0.665 at 0.762). Further controls rule out encoder scale, aggregation choice, and two pre-specified lexical mechanisms as explanations (Appendices I and K). CoVeR’s benefit therefore depends on identifying specific skippable states rather than on matching an overall skip rate. The precise source of its advantage over lexical overlap remains unresolved.

5.4 Transfer: Deciders, Agent Scale, and LLM-Free Routing

A public prompt judge as an independent decider. The trained verifier is one instantiation; the gate also routes a fully public decider. Replacing the in-band decider with a zero-shot prompt-only sufficiency judge and keeping the *same* threshold with no re-tuning, all 6 cells (3 datasets $\times$ 2 judge thresholds) land within 1 pp of the always-verify judge at $\geq 40\%$ judge-call cut (cuts 56.7–95.2%). The judge’s own absolute EM is lower than the trained verifier’s, but it is a fully public decider whose accuracy the gate preserves while removing most of its calls. A third decider — a public, off-the-shelf NLI cross-encoder (DeBERTa-v3 MNLI, not fine-tuned by us) over declarative forms of the same claims — shows the same routing behavior, removing 61–68% of its calls on HotpotQA/2Wiki and 94–95% on MuSiQue (App. F); its accuracy effect is discussed with the other correction evidence in §6.

A larger agent. The coverage geometry survives the scale change: on Self-Ask 7B trajectories the oracle-target AUROC is 0.872/0.866/0.869, within ~ 0.04 of 3B, while the deployable generated-claim margin remains weak (0.632/0.572/0.711). Transferring the same threshold onto Self-Ask 7B trajectories, now with the verifier on generated expected claims (the fully deployable setting), again removes about two thirds of verifier calls while EM stays within a fraction of a point of the always-verify baseline on every dataset (App. F, panel a). The behavior is not specific to the 3B agent, and it holds without gold labels. Generated claims are harder to satisfy than gold supporting facts, so the deployable verifier fires far less often at this scale (51.0 $\rightarrow$ 19.2 / 36.6 $\rightarrow$ 14.0 / 23.8 $\rightarrow$ 2.6%). The lower firing rate creates more opportunities to skip calls whose verdict would not have altered the control action.

Routing without an LLM. Replacing claim generation *for the margin* with the sub-1M distilled drafter (CoVeR-D) routes end-to-end and preserves the generated-claim gate’s EM at a comparable call cut (App. F, panel b) — the distilled gate preserves end-to-end answer quality, not only per-claim discrimination. Pairing that drafter with the claim-free prompt judge removes claim generation from the *whole* pipeline, and the gate still routes the judge within 1 pp of always-verify at 56–92% judge-call cut (App. L).

Beyond the closed pool. The main results use a closed distractor pool. The same threshold transfers to an open-corpus setting (HotpotQA Setting-B, Self-Ask 3B over E5-dense Wikipedia-2018) and cuts 92.1% of verifier calls at EM indistinguishable from both references (+0.30 pp each) and 0.97 stop-agreement. This is a saturation regime (gold verifier fire 6.4%) run with the gold-claim decider, so it is a single disclosed point rather than a second confirmatory setting — but the threshold does carry out of the distractor pool without re-tuning. All three transfer arms are tabulated in App. F.

5.5 Verification Workload

Two tiers, two regimes. We report verifier workload at three levels — whole decider calls, claim-level evaluations, and input tokens — and account for the middle level as claim-verdict *pairs* — one claim adjudicated once — per question (Figure 4; per-stage counts in App. C). Tier 1, call routing, removes whole decider calls and holds whether the decider is batched or not, eliminating 62–68% of pairs on HotpotQA and 2Wiki and 93% on MuSiQue. Tier 2, intra-call ordering, removes claims within a retained call; it forgoes intra-call

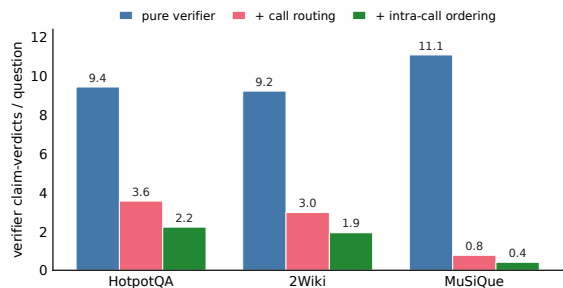


Figure 4: **Verification workload by tier.** Verifier claim-verdicts per question at each routing tier. Tier 1 holds batched; tier 2 saves only unbatched.

batching, so its saving arises only under sequential scoring. Stacked, the two tiers remove 76–96% of pairs.

Workload and latency scope. The claim is a reduction in verifier *workload*, in deployment-portable units: 62–68% fewer decider calls, 62–68% fewer claim-verdict pairs, and 72/73/97% fewer verifier input tokens (a GPU replay confirms the token and wall-clock reductions; App. C). Gating also defers stops, adding at most half a retrieval loop per question vs. always-verify (App. C); whether the saved verifier work outweighs the added loops depends on the relative cost of a verifier call and a generation loop in a given deployment — we make no deployment-independent end-to-end latency claim, and give a break-even rule plus one stack-specific timing study in App. C. (*Drafter cost.*) The generated-claim gate spends one claim-generation LLM pass per question — a pass always-verify also pays; the distilled drafter replaces it for the margin computation (a sub-1M MLP forward).

The margin can also route the symmetric resume decision for native-stopped trajectories (bounded positive; App. D). In a new deployment: embed the claims the agent already generates, select the threshold on a small training split (§3), and route; the same LODO-selected $\tau=0.16$ transferred across deciders and agent scales, as well as to the open-corpus setting.

6 Analysis: When Can the Routing Signal Correct the Decider?

Routing and correction impose different requirements on the cheap signal: routing needs only calibration (§2), while correction requires error decorrelation (Madras, Pitassi, and Zemel 2018; Mozannar and Sontag 2020). Our two deciders instantiate the two sides of that condition.

6.1 Correlated Errors Preclude Correction: the Trained Verifier

We partition questions into three paths: retained decider-call stops, budget-exhausted runs, and skip-diverged trajectories. Decomposing the gate’s $\Delta\text{EM-vs-Full}$ by path (primary arm, full n), skip-diverged trajectories differ by at most one question and the budget path equals Full up to re-extraction flips; the substantive residual is concentrated in the *decider-stop*

path and it is inherited from always-verify *exactly* ($\Delta\text{EM-vs-always} = 0$ on that path): early wrong stops fire in the first loops and match a wrong-entity page more often than correct stops do (on 2Wiki the same path inherits a small gain). So can the margin veto *those* mistakes?

A pre-specified veto gate (evaluated on the $n=400$ pilot arm) asks whether the margin-at-fire (or the verifier’s own match-margin) ranks error fires above correct fires at AUROC ≥ 0.65 . It fails: AUROC 0.39 on the decider-stop path, 0.36 over all fires, CIs including chance 0.5 — *no recoverable veto signal* (primary-arm error counts stay small, 14/6/1). Only a trivial fire-loop control separates (0.76), largely a skip-first- k position heuristic. The reason is mechanistic: margin and verifier are *both* driven by claim–evidence similarity, so evidence deceptive enough to fool the verifier also looks covered to the margin — their errors correlate, exactly the non-complementary regime where deferral theory predicts no gain; the inherited decider-stop residual remains.

6.2 Decorrelated Errors May Permit It: the Volatile Judge (Exploratory)

The prompt judge errs differently: its per-loop p_{yes} is volatile (mean absolute loop-to-loop change 0.09–0.16), so its early “sufficient” calls are unreliable *and* margin-separable, and the gate vetoes early stops at high-margin loops. Overrides are positive but underpowered: pooled +1.77 pp at $\tau_{\text{judge}}=0.5$ (CI spans zero); similar at 0.95. The public NLI decider (§5.4) shows the same override pattern, and for that decider the effect is measurable: on HotpotQA the gated decider’s ΔEM is strictly *positive* (+2.8 pp, CI>0) because the gate vetoes premature stops, so its deviation from always-verify is favorable rather than an accuracy loss. We treat the judge result as consistent with, not establishing, decorrelation. What it separates cleanly is the two deciders: the judge’s errors are visible to the margin, the trained verifier’s are not—the learning-to-defer complementarity condition.

7 Conclusion

CoVeR routes verifier calls using a single threshold on a frozen coverage margin, while every stopping decision is still issued by the decider. Across three multi-hop QA benchmarks it preserves answer accuracy relative to both references while reducing verifier calls by 62–68% in the representative regimes. The same threshold, never tuned on the dataset it is evaluated on, transfers across deciders and agent scales, and a sub-1M drafter removes LLM computation from the routing path. The margin cannot replace verification, nor correct failures it shares with the decider, which indicates that correction would require a signal whose errors decorrelate from the decider’s.

Limitations. Our evidence is one agent scaffold, three multi-hop QA datasets, and a single saturated open-corpus point. Absolute EM is low, so sub-1 pp effects are unresolvable and the HotpotQA CI does not exclude a 1 pp loss. The gate inherits the decider’s early-stop errors (§6), trajectories are single-seed, and the claim is a workload reduction, not an end-to-end one (App. C; further disclosures §4, App. B).

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